Protein Analysis Report

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Bioinformatics Analysis Report: Protein Structural and Compositional Trends

Theoretical Introduction

The analysis of protein data provides crucial insights into both the evolution of biological mechanisms and the advancement of structural biology techniques. This report investigates two main areas: the historical progression of structure determination methods and the statistical composition of proteins across different organisms.

Historically, X-ray crystallography has dominated the Protein Data Bank (PDB). However, X-ray crystallography requires a challenging, time-consuming, and highly expensive protein crystallization process. In contrast, recent technological leaps in detector technology and computational processing have caused a "Resolution Revolution" in Cryo-Electron Microscopy (Cryo-EM)[cite: 1]. Cryo-EM allows for the study of large macromolecular complexes in their native, aqueous state without the need for crystals, fundamentally shifting how structures are solved and drastically reducing historical bottlenecks.

Biologically, the primary structure of a protein dictates more than just its fold; it governs its cellular lifespan. According to the **N-end rule**, the *in vivo* half-life of a protein is heavily influenced by its N-terminal amino acid residue[cite: 2]. Destabilizing N-terminal residues trigger rapid ubiquitin-dependent (in eukaryotes) or distinct proteolytic degradation (in bacteria). Consequently, statistical analysis of N-termini and overall amino acid content across diverse species (E coli, Human, Yeast, A thaliana, D melanogaster, C elegans, Mouse, Zebrafish, B subtilis) and broad taxonomic kingdoms (Bacteria, Viruses, Archaea, Eukaryota) can reveal evolutionary pressures and distinct metabolic strategies employed by different organisms.

Methodology

Structural Database Trends Analysis (PDB)

Metadata was acquired directly from the PDB (`entries.idx` and `pdb_seqres.txt`). By filtering specifically for protein entries and weighting the data using the formula $\sum(n \cdot 1/r)$ (where n is the sequence length, and r is resolution). This solution took into consideration not only length but also quality of data.

Protein Length and Composition Analysis

Sequences for targeted species (E coli, Human, Yeast, A thaliana, D melanogaster, C elegans, Mouse, Zebrafish, B subtilis) and randomly sampled sequences from kingdoms (100 sequences each) were retrieved from the UniProtKB REST API. Random sampling of kingdoms partially prevents database bias from skewing the macroscopic view. To estimate the standard error of mean protein lengths and amino acid frequencies, statistical bootstrapping (95% confidence interval) was employed.

Results and Analysis

Structural Database Trends Analysis (PDB)

Structural Analysis, Interactive Data

Note: The interactive structural growth timeline (2005-2025) is available in the HTML version of this report.

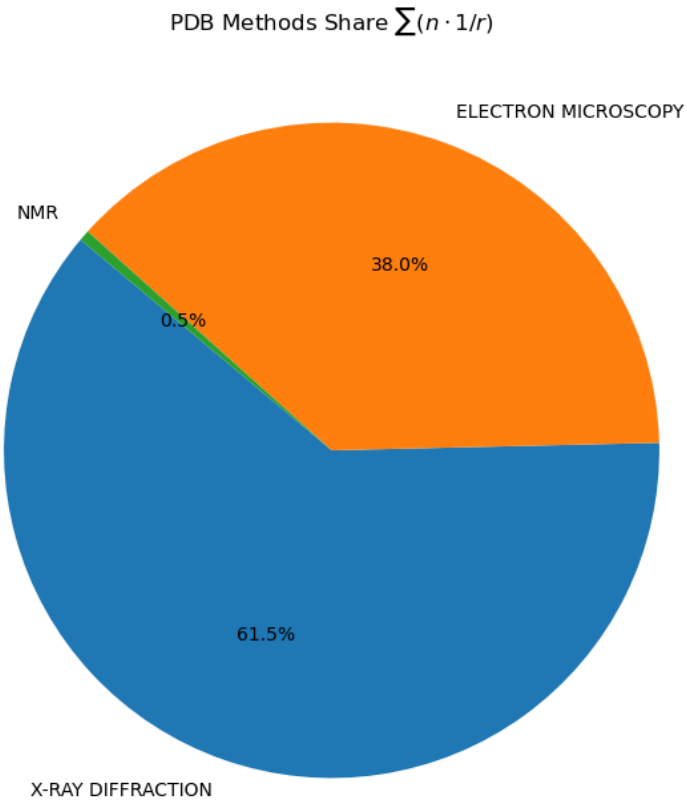


Figure 1: Overall distribution of structural determination methods in the PDB.

We can observe that in recent years *electron microscopy* became more popular. That is because it provides general method for investigating proteins with high resolution. It also does not incorporate protein crystallization which is hardest part in X ray crystallography.

Protein Length and Composition Analysis

Based on the parsed UniProt FASTA files, we analyzed the frequency of specific amino acids at the N-terminus, as well as the overall sequence length distributions.

Length comparison between groups

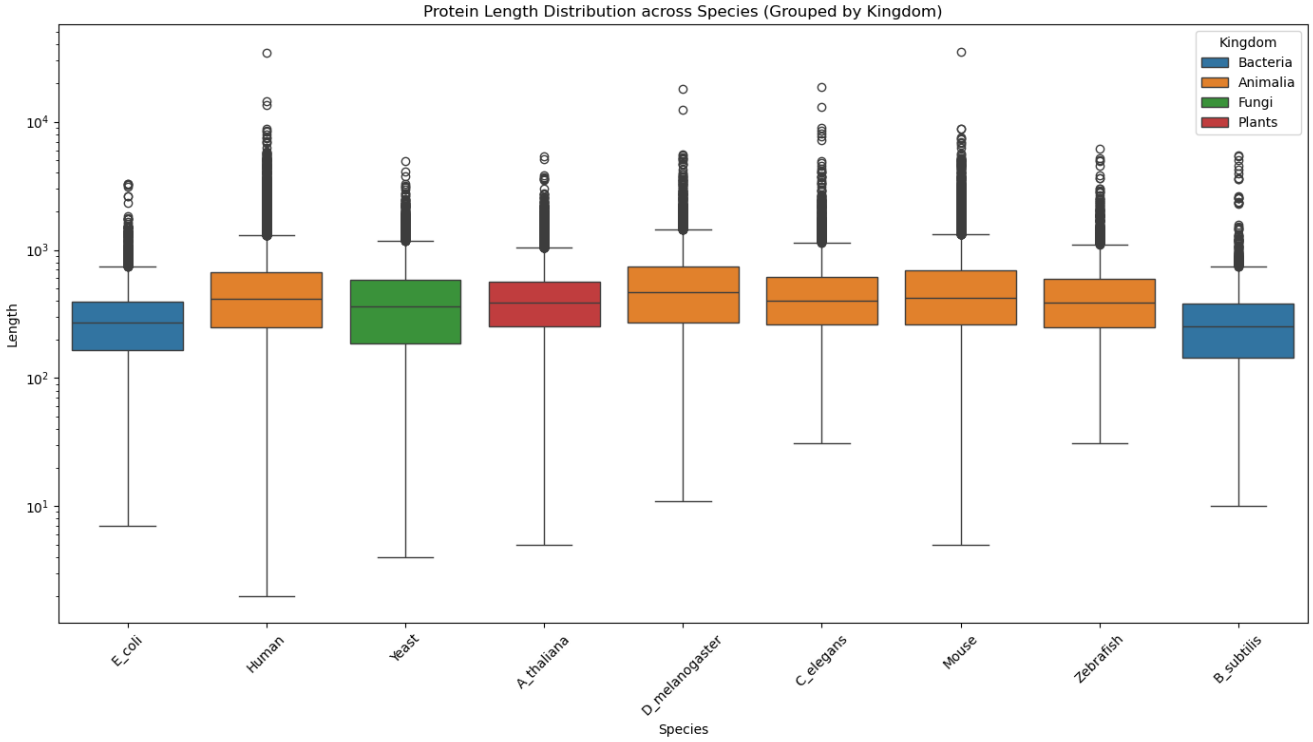


Figure 2: Distribution of protein lengths across selected species grouped by taxonomy.

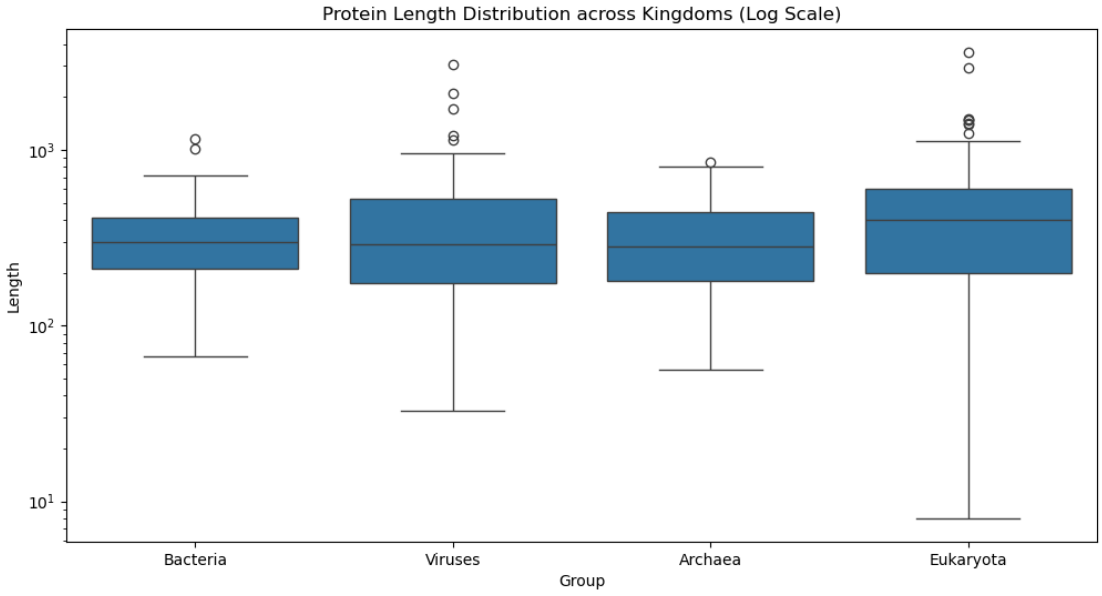


Figure 3: Distribution of protein lengths across kingdoms.

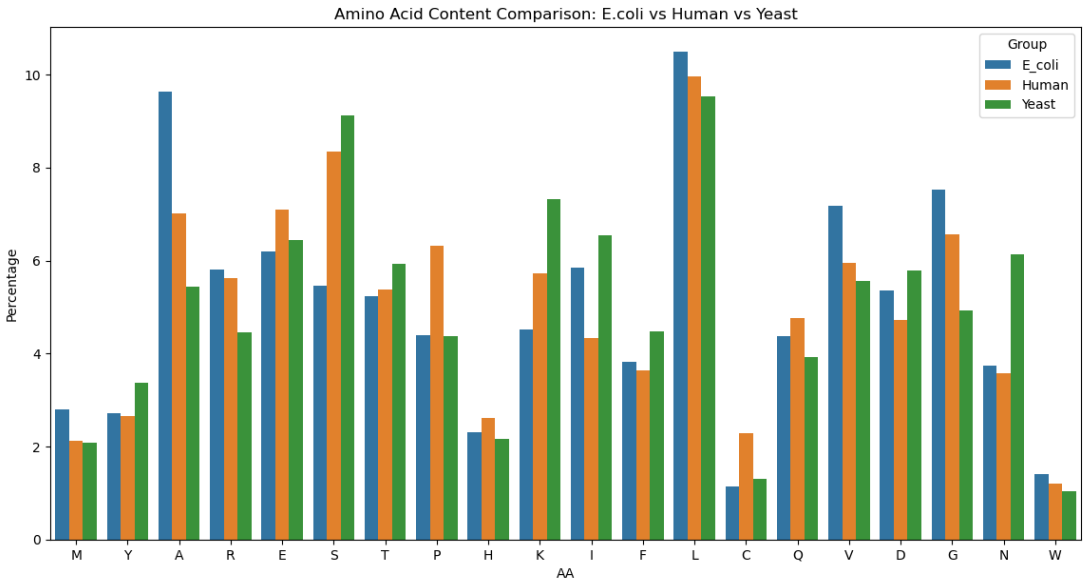


Figure 4: Percentage content of amino acids across E.coli, Human, and Yeast.

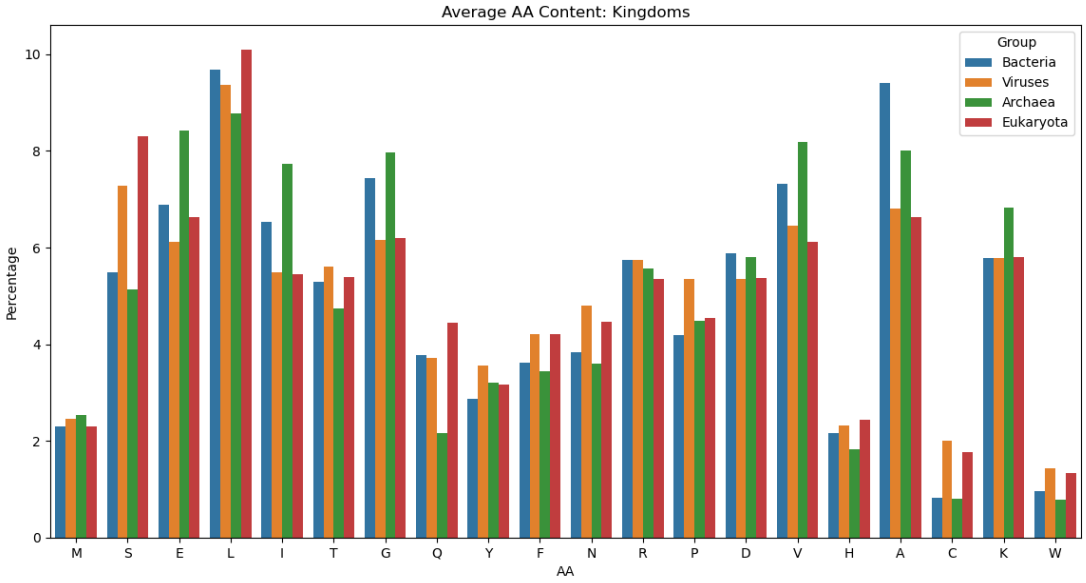
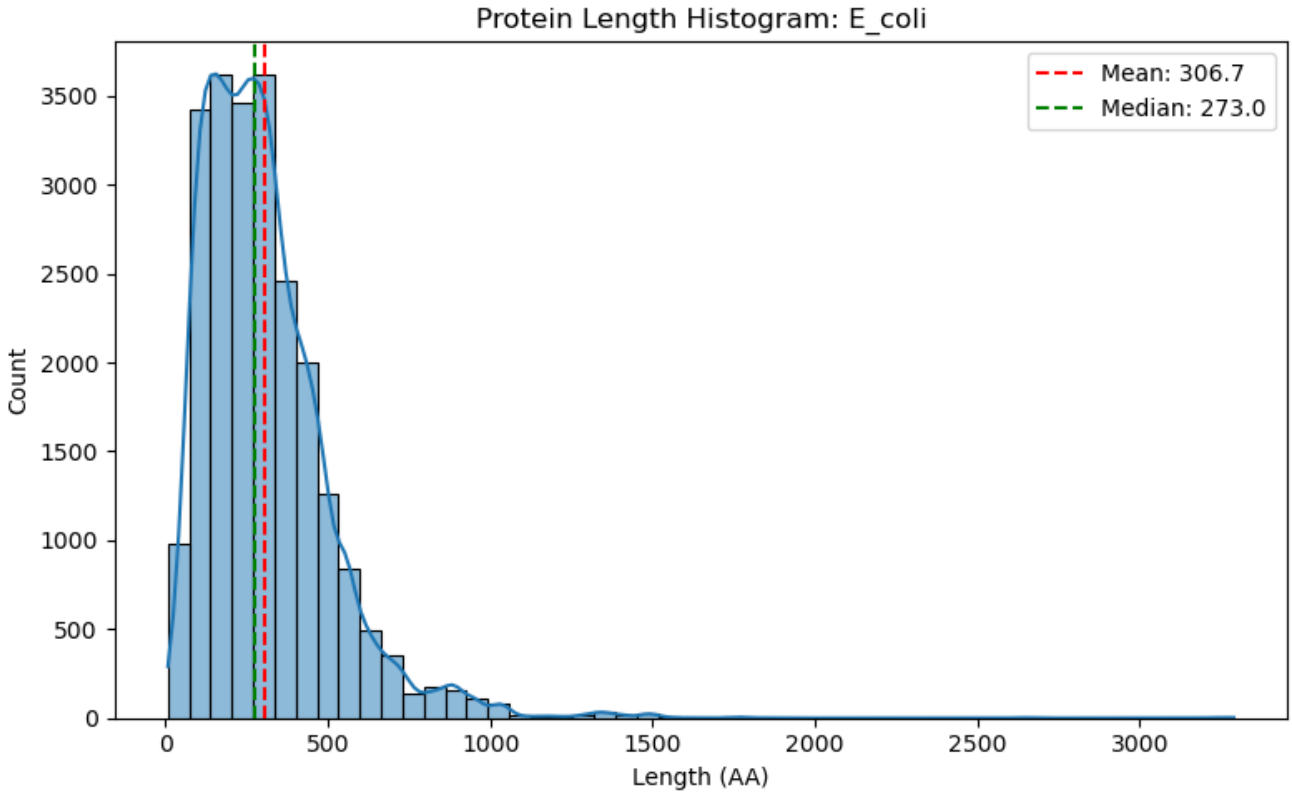
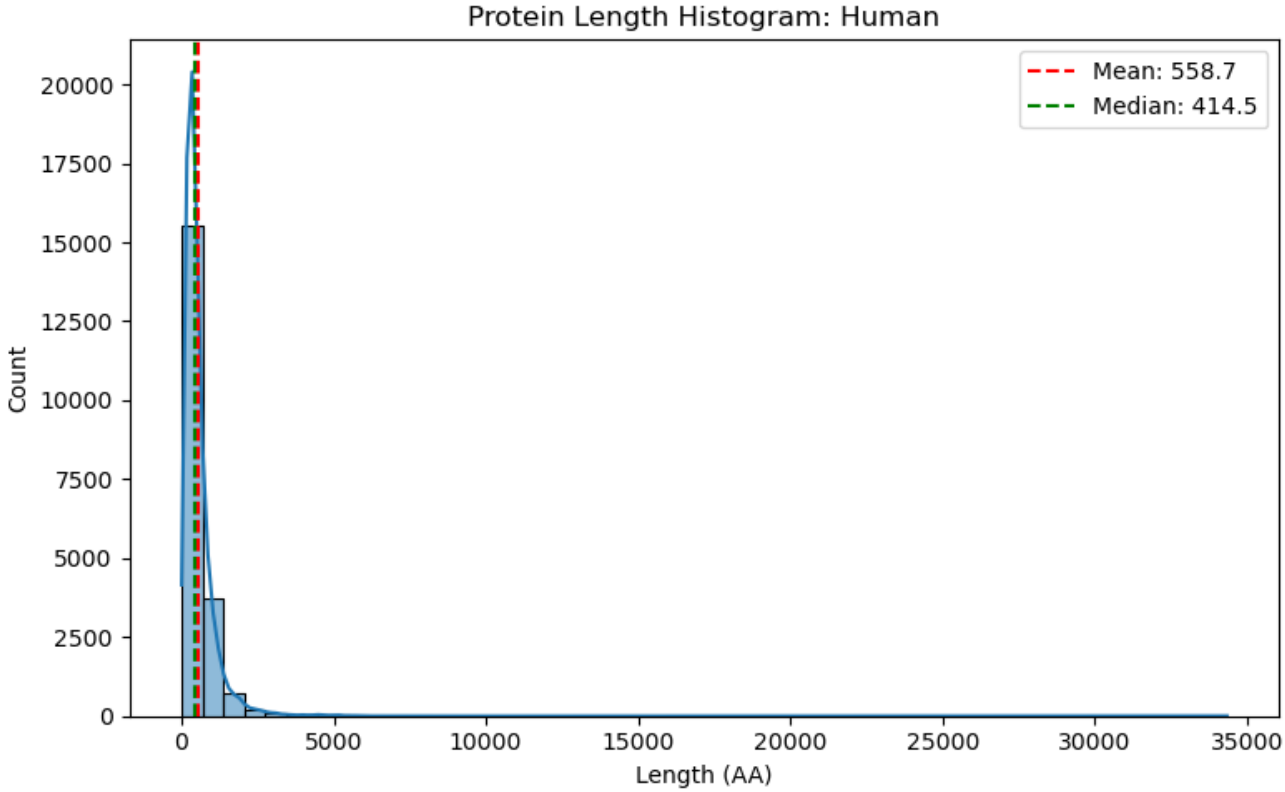
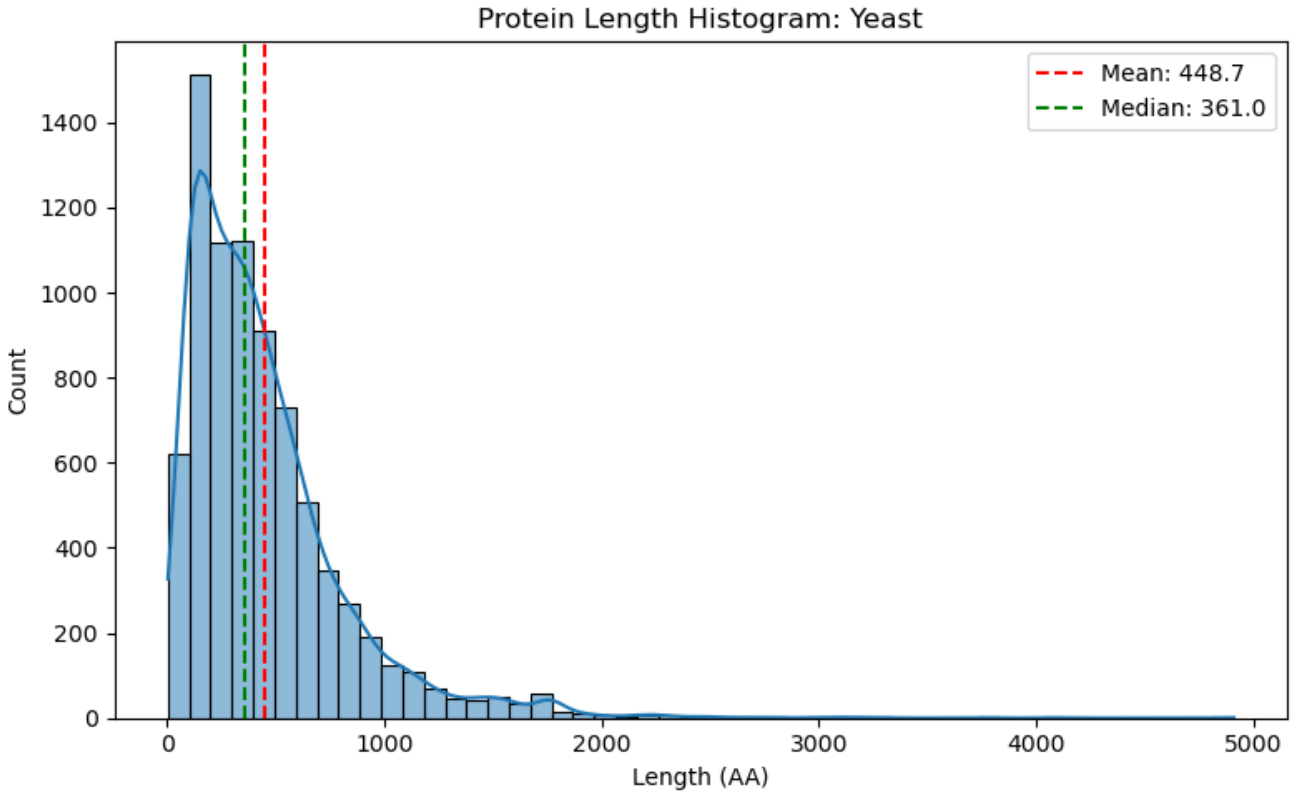


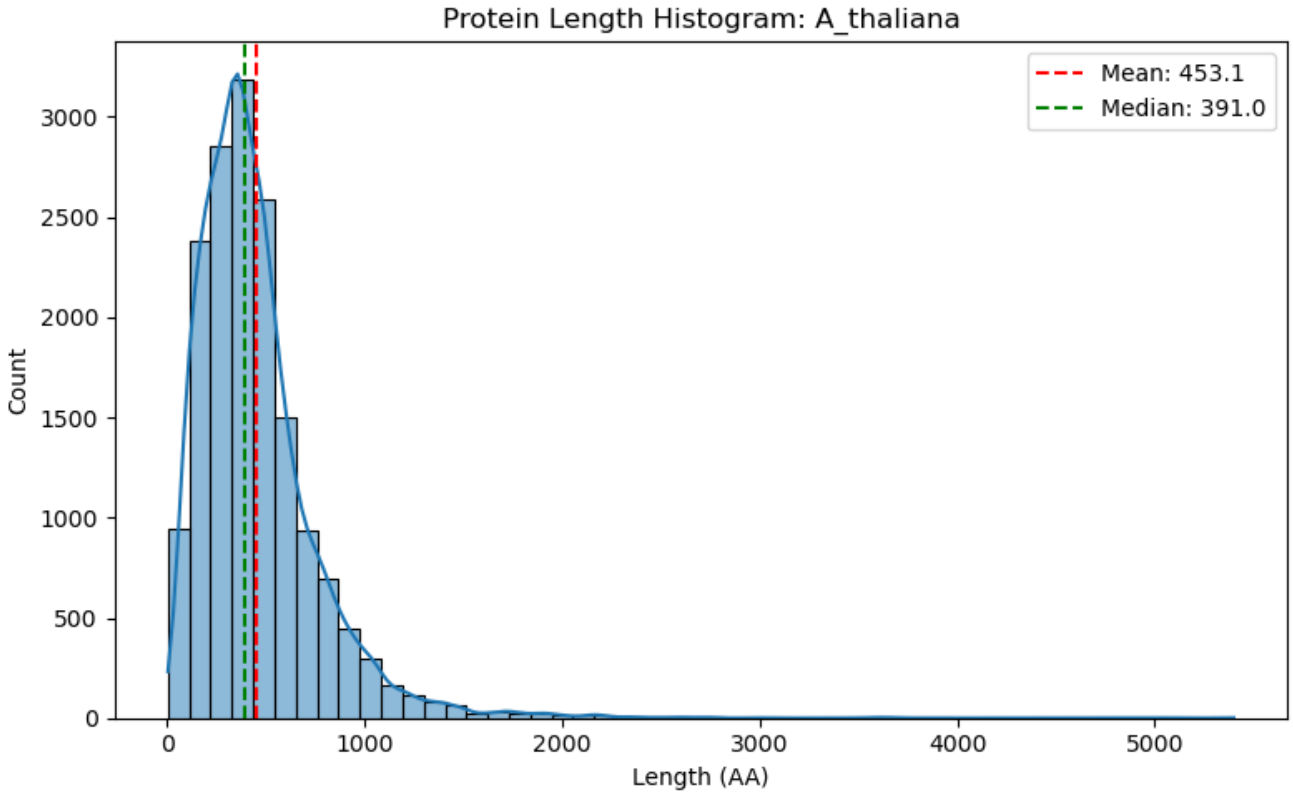
Figure 5: Average amino acid content across taxonomic kingdoms.

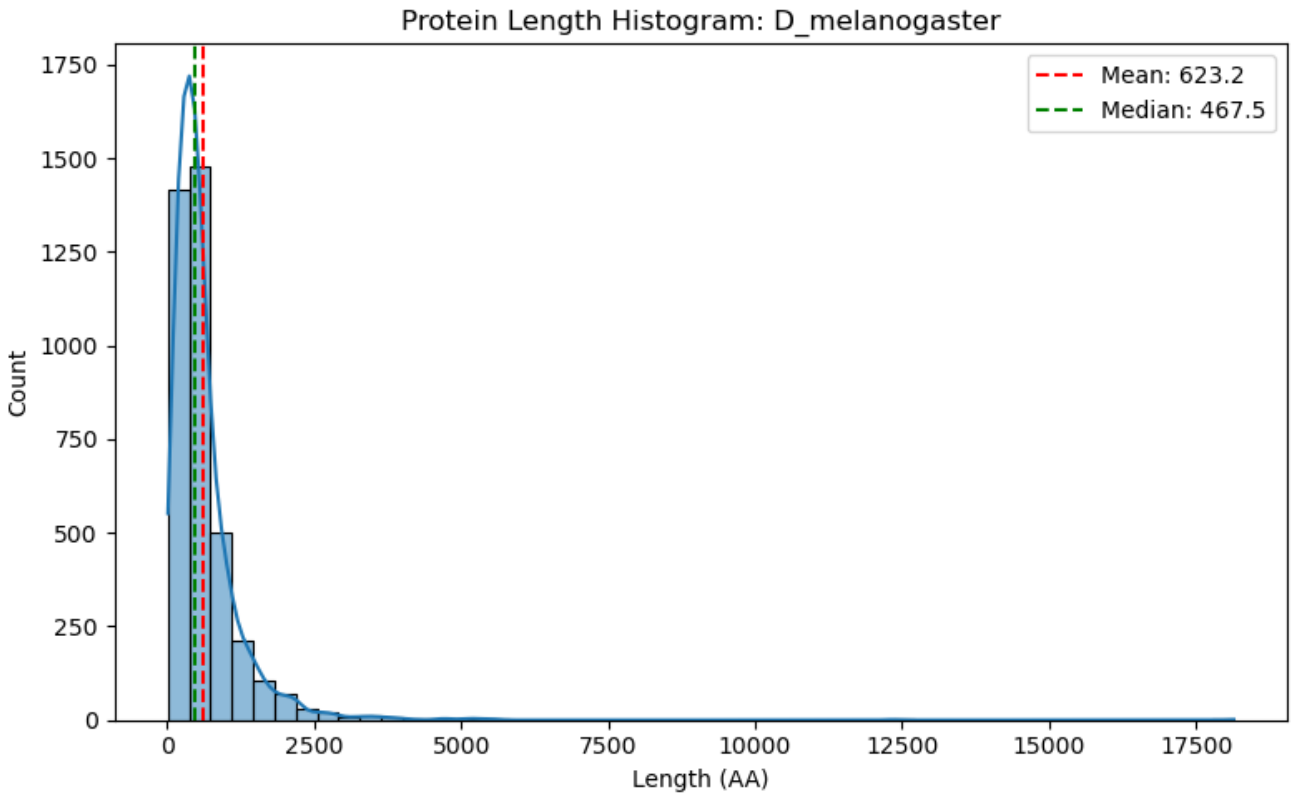
Protein Length Histograms

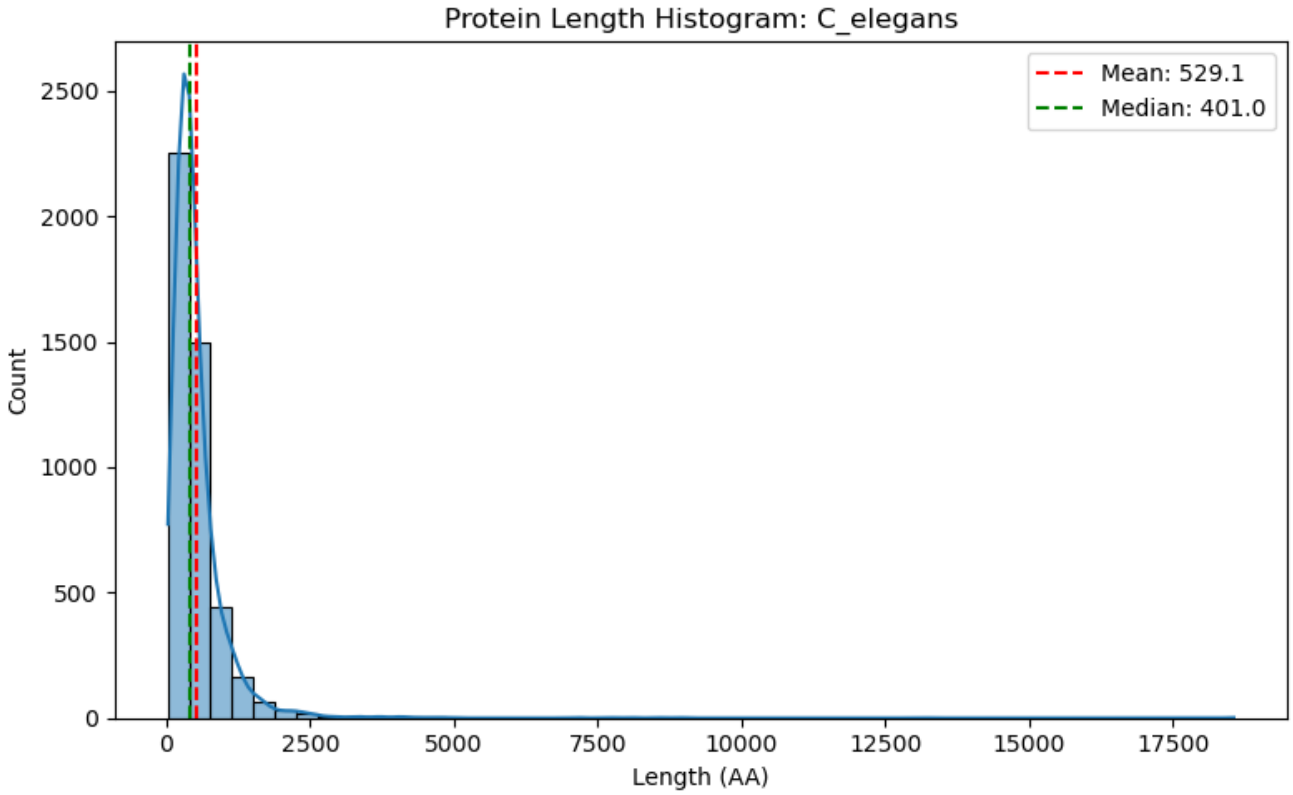

Protein Length Histogram: E_coli
Mean: 306.7
Median: 273.0
Count
Length (AA)
0
500
1000
1500
2000
2500
3000
3500

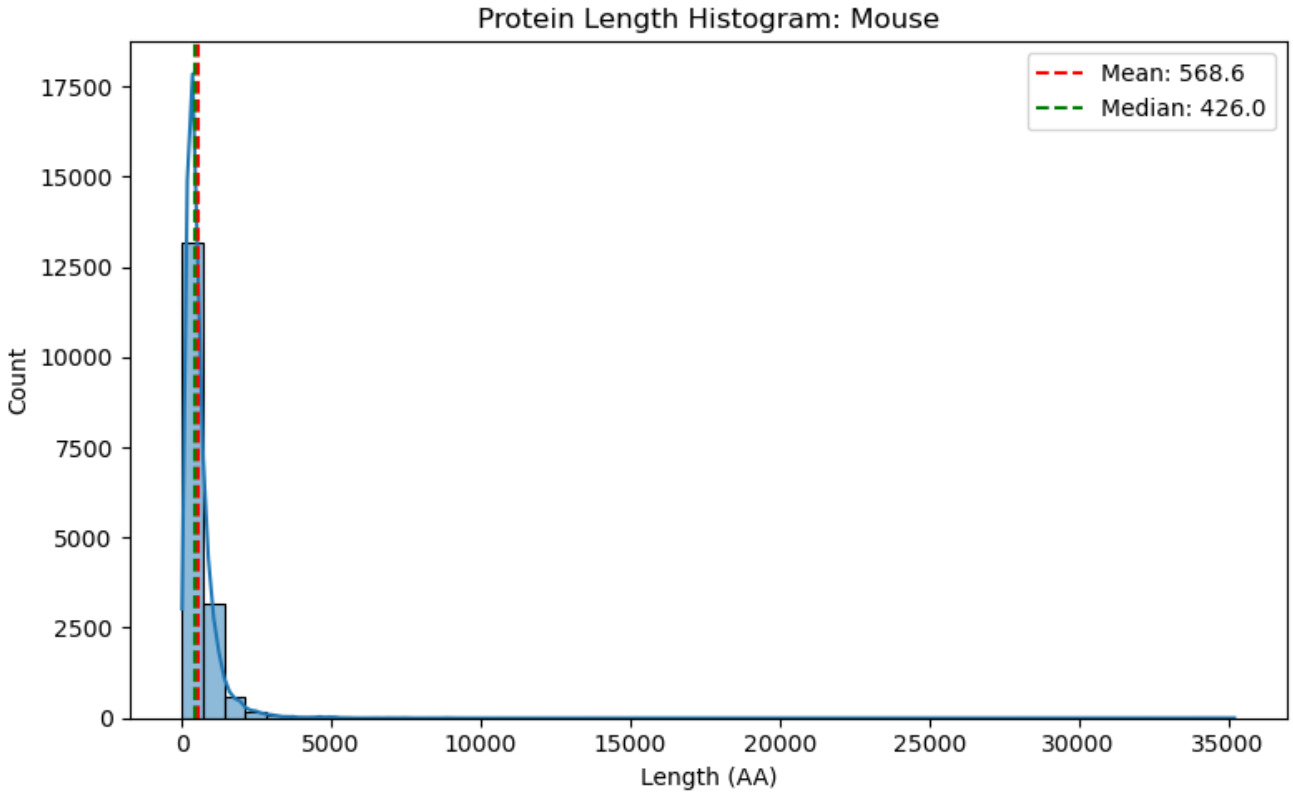

Protein Length Histogram: Human
Mean: 558.7
Median: 414.5
Count
Length (AA)
0
2500
5000
7500
10000
12500
15000
17500
20000
25000
30000
35000

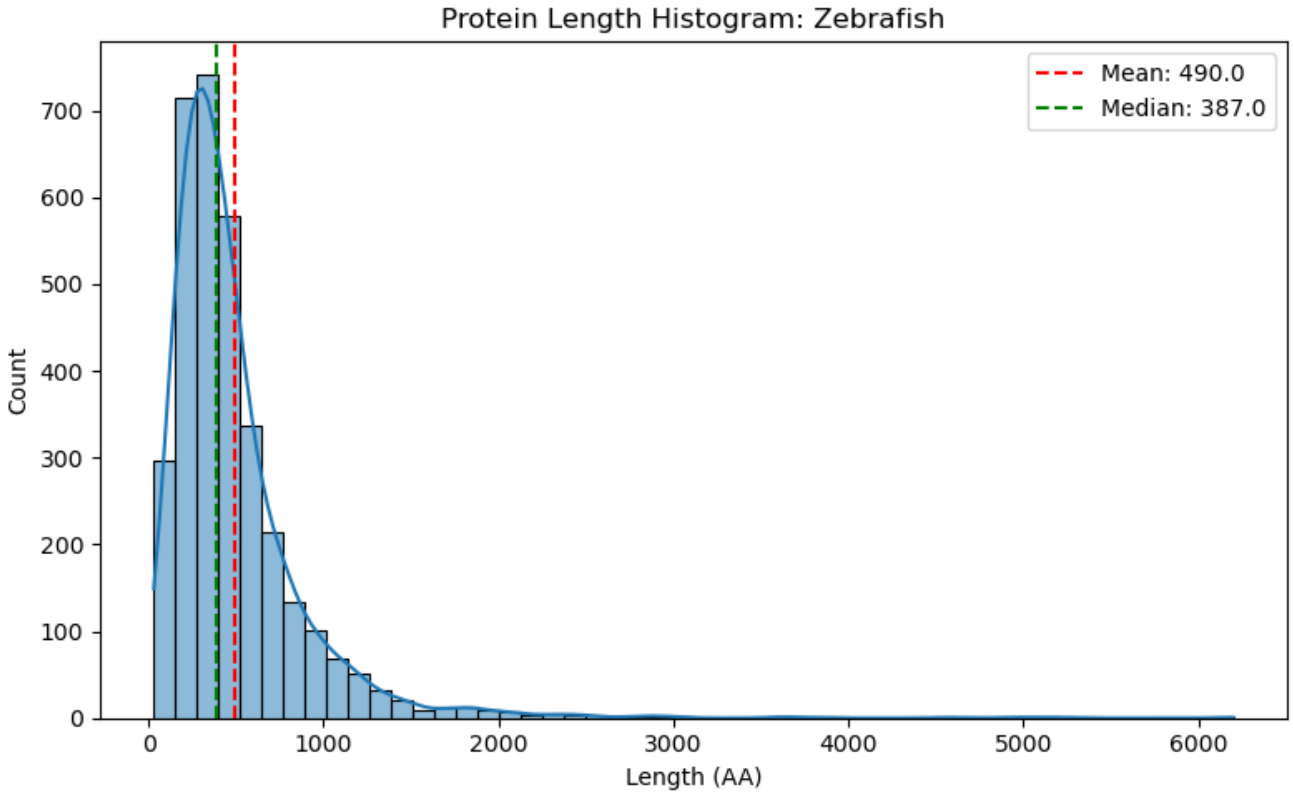

Protein Length Histogram: Yeast
Mean: 448.7
Median: 361.0
Count
Length (AA)
0
200
400
600
800
1000
1200
1400
0
1000
2000
3000
4000
5000

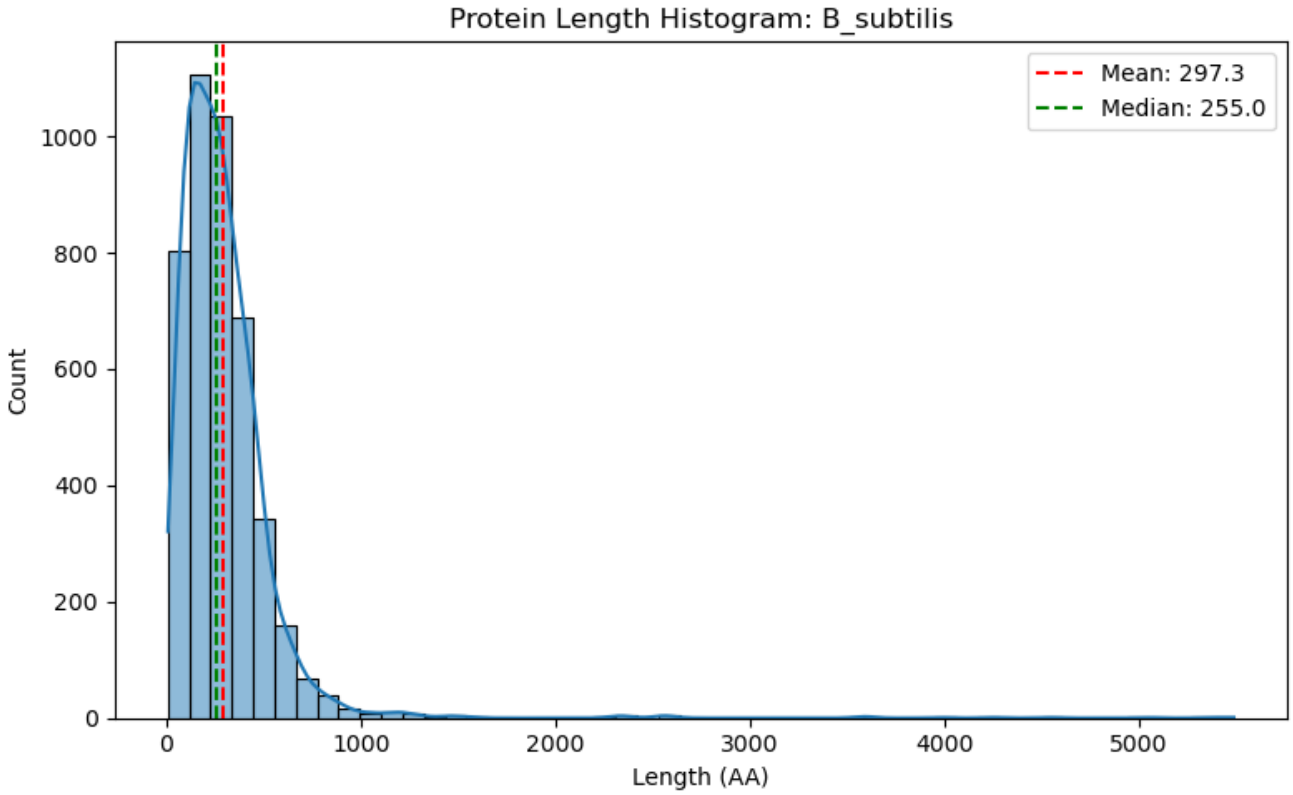

Protein Length Histogram: A_thaliana
Mean: 453.1
Median: 391.0
Count
Length (AA)
0
500
1000
1500
2000
2500
3000
0
1000
2000
3000
4000
5000


Protein Length Histogram: D_melanogaster
Mean: 623.2
Median: 467.5
Count
Length (AA)
0
250
500
750
1000
1250
1500
1750
0
2500
5000
7500
10000
12500
15000
17500


Protein Length Histogram: C_elegans
Mean: 529.1
Median: 401.0
Count
Length (AA)
0
500
1000
1500
2000
2500
0
2500
5000
7500
10000
12500
15000
17500


Protein Length Histogram: Mouse
Mean: 568.6
Median: 426.0
Count
Length (AA)
0
2500
5000
7500
10000
12500
15000
17500
0
5000
10000
15000
20000
25000
30000
35000


Protein Length Histogram: Zebrafish
Mean: 490.0
Median: 387.0
Count
Length (AA)
0
100
200
300
400
500
600
700
0
1000
2000
3000
4000
5000
6000



Statistical Tables

Species - Length Statistics

Group	Mean_Length	SE	Median_Length	Sample_Size
E_coli	306.71	2.74	273	23290
Human	558.67	7.53	414	20434
Yeast	448.67	5.18	361	7933
A_thaliana	453.07	4.23	391	16418
D_melanogaster	623.24	10.56	467	3874
C_elegans	529.14	8.71	401	4499
Mouse	568.64	7.84	426	17274
Zebrafish	489.95	6.95	387	3369
B_subtilis	297.32	4.3	255	4312

Species - Amino Acid Content (with StdErr)

Group	AminoAcid	Percentage	StdErr
E_coli	A	9.64	0.193
E_coli	C	1.14	0.069
E_coli	D	5.37	0.148
E_coli	E	6.19	0.158

Group	AminoAcid	Percentage	StdErr
E_coli	F	3.83	0.126
E_coli	G	7.52	0.173
E_coli	H	2.31	0.098
E_coli	I	5.85	0.154
E_coli	K	4.52	0.136
E_coli	L	10.5	0.201
E_coli	M	2.8	0.108
E_coli	N	3.75	0.124
E_coli	P	4.41	0.134
E_coli	Q	4.37	0.134
E_coli	R	5.8	0.153
E_coli	S	5.47	0.149
E_coli	T	5.24	0.146
E_coli	V	7.18	0.169
E_coli	W	1.41	0.077
E_coli	Y	2.71	0.106
Human	A	7.01	0.179
Human	C	2.3	0.105
Human	D	4.73	0.149
Human	E	7.11	0.18
Human	F	3.65	0.131
Human	G	6.56	0.173
Human	H	2.62	0.112
Human	I	4.34	0.143
Human	K	5.73	0.163
Human	L	9.96	0.209
Human	M	2.13	0.101
Human	N	3.58	0.13
Human	P	6.32	0.17
Human	Q	4.77	0.149
Human	R	5.63	0.161
Human	S	8.35	0.194
Human	T	5.38	0.158
Human	V	5.96	0.166
Human	W	1.21	0.077
Human	Y	2.66	0.113
Yeast	A	5.45	0.255
Yeast	C	1.31	0.127
Yeast	D	5.78	0.262
Yeast	E	6.45	0.276
Yeast	F	4.48	0.232
Yeast	G	4.94	0.243
Yeast	H	2.18	0.164
Yeast	I	6.54	0.278
Yeast	K	7.33	0.293
Yeast	L	9.53	0.33
Yeast	M	2.09	0.161
Yeast	N	6.14	0.27

Group	AminoAcid	Percentage	StdErr
Yeast	P	4.38	0.23
Yeast	Q	3.92	0.218
Yeast	R	4.47	0.232
Yeast	S	9.12	0.323
Yeast	T	5.93	0.265
Yeast	V	5.56	0.257
Yeast	W	1.04	0.114
Yeast	Y	3.37	0.203
A_thaliana	A	6.48	0.192
A_thaliana	C	1.8	0.104
A_thaliana	D	5.34	0.175
A_thaliana	E	6.6	0.194
A_thaliana	F	4.34	0.159
A_thaliana	G	6.55	0.193
A_thaliana	H	2.22	0.115
A_thaliana	I	5.42	0.177
A_thaliana	K	6.26	0.189
A_thaliana	L	9.64	0.23
A_thaliana	M	2.5	0.122
A_thaliana	N	4.41	0.16
A_thaliana	P	4.74	0.166
A_thaliana	Q	3.47	0.143
A_thaliana	R	5.3	0.175
A_thaliana	S	8.88	0.222
A_thaliana	T	5.13	0.172
A_thaliana	V	6.77	0.196
A_thaliana	W	1.25	0.087
A_thaliana	Y	2.9	0.131
D_melanogaster	A	7.42	0.421
D_melanogaster	C	1.8	0.214
D_melanogaster	D	5.27	0.359
D_melanogaster	E	6.49	0.396
D_melanogaster	F	3.58	0.298
D_melanogaster	G	6.09	0.384
D_melanogaster	H	2.66	0.258
D_melanogaster	I	4.98	0.349
D_melanogaster	K	5.69	0.372
D_melanogaster	L	9.27	0.466
D_melanogaster	M	2.39	0.245
D_melanogaster	N	4.71	0.34
D_melanogaster	P	5.28	0.359
D_melanogaster	Q	5.3	0.36
D_melanogaster	R	5.46	0.365
D_melanogaster	S	8.2	0.441
D_melanogaster	T	5.57	0.368
D_melanogaster	V	5.91	0.379
D_melanogaster	W	0.99	0.159
D_melanogaster	Y	2.94	0.272

Group	AminoAcid	Percentage	StdErr
C_elegans	A	6.66	0.372
C_elegans	C	1.88	0.203
C_elegans	D	5.69	0.345
C_elegans	E	6.92	0.378
C_elegans	F	4.24	0.3
C_elegans	G	5.55	0.341
C_elegans	H	2.36	0.226
C_elegans	I	5.76	0.347
C_elegans	K	6.37	0.364
C_elegans	L	8.43	0.414
C_elegans	M	2.66	0.24
C_elegans	N	4.76	0.318
C_elegans	P	4.97	0.324
C_elegans	Q	4.28	0.302
C_elegans	R	5.42	0.338
C_elegans	S	7.99	0.404
C_elegans	T	5.72	0.346
C_elegans	V	6.26	0.361
C_elegans	W	1.08	0.154
C_elegans	Y	3.02	0.255
Mouse	A	6.99	0.194
Mouse	C	2.22	0.112
Mouse	D	4.88	0.164
Mouse	E	7.05	0.195
Mouse	F	3.67	0.143
Mouse	G	6.51	0.188
Mouse	H	2.56	0.12
Mouse	I	4.31	0.155
Mouse	K	5.65	0.176
Mouse	L	10.05	0.229
Mouse	M	2.16	0.111
Mouse	N	3.56	0.141
Mouse	P	6.2	0.184
Mouse	Q	4.77	0.162
Mouse	R	5.62	0.175
Mouse	S	8.42	0.211
Mouse	T	5.36	0.171
Mouse	V	6.12	0.182
Mouse	W	1.22	0.083
Mouse	Y	2.68	0.123
Zebrafish	A	6.55	0.426
Zebrafish	C	2.22	0.254
Zebrafish	D	5.27	0.385
Zebrafish	E	6.99	0.439
Zebrafish	F	3.74	0.327
Zebrafish	G	6.01	0.41
Zebrafish	H	2.68	0.278
Zebrafish	I	4.55	0.359

Group	AminoAcid	Percentage	StdErr
Zebrafish	K	5.84	0.404
Zebrafish	L	9.8	0.512
Zebrafish	M	2.52	0.27
Zebrafish	N	3.87	0.332
Zebrafish	P	5.35	0.388
Zebrafish	Q	4.77	0.367
Zebrafish	R	5.64	0.397
Zebrafish	S	8.52	0.481
Zebrafish	T	5.4	0.389
Zebrafish	V	6.3	0.419
Zebrafish	W	1.17	0.185
Zebrafish	Y	2.82	0.285
B_subtilis	A	7.64	0.404
B_subtilis	C	0.78	0.134
B_subtilis	D	5.21	0.338
B_subtilis	E	7.25	0.395
B_subtilis	F	4.5	0.316
B_subtilis	G	6.89	0.386
B_subtilis	H	2.28	0.227
B_subtilis	I	7.36	0.398
B_subtilis	K	7.07	0.39
B_subtilis	L	9.64	0.449
B_subtilis	M	2.77	0.25
B_subtilis	N	4	0.298
B_subtilis	P	3.67	0.286
B_subtilis	Q	3.85	0.293
B_subtilis	R	4.08	0.301
B_subtilis	S	6.29	0.37
B_subtilis	T	5.43	0.345
B_subtilis	V	6.72	0.381
B_subtilis	W	1.04	0.154
B_subtilis	Y	3.53	0.281

Kingdom - Length Statistics

Group	Mean_Length	SE	Median_Length	Sample_Size
Bacteria	325.04	17.97	299	100
Viruses	416.99	41.71	292	100
Archaea	313.95	18.29	281	100
Eukaryota	509.43	50.24	398	100

Kingdom - Amino Acid Content (with StdErr)

Group	AminoAcid	Percentage	StdErr
Bacteria	A	9.41	2.92

Group	AminoAcid	Percentage	StdErr
Bacteria	C	0.83	0.906
Bacteria	D	5.88	2.352
Bacteria	E	6.88	2.531
Bacteria	F	3.62	1.869
Bacteria	G	7.44	2.624
Bacteria	H	2.17	1.458
Bacteria	I	6.53	2.47
Bacteria	K	5.78	2.333
Bacteria	L	9.68	2.958
Bacteria	M	2.3	1.498
Bacteria	N	3.83	1.92
Bacteria	P	4.2	2.006
Bacteria	Q	3.77	1.905
Bacteria	R	5.75	2.328
Bacteria	S	5.49	2.279
Bacteria	T	5.29	2.238
Bacteria	V	7.32	2.604
Bacteria	W	0.96	0.977
Bacteria	Y	2.87	1.669
Viruses	A	6.8	2.518
Viruses	C	2	1.4
Viruses	D	5.35	2.249
Viruses	E	6.12	2.397
Viruses	F	4.2	2.006
Viruses	G	6.16	2.405
Viruses	H	2.33	1.509
Viruses	I	5.48	2.276
Viruses	K	5.78	2.333
Viruses	L	9.36	2.913
Viruses	M	2.46	1.548
Viruses	N	4.79	2.136
Viruses	P	5.36	2.251
Viruses	Q	3.71	1.891
Viruses	R	5.75	2.328
Viruses	S	7.28	2.599
Viruses	T	5.6	2.3
Viruses	V	6.46	2.459
Viruses	W	1.43	1.188
Viruses	Y	3.57	1.854
Archaea	A	8.01	2.715
Archaea	C	0.81	0.894
Archaea	D	5.81	2.339
Archaea	E	8.42	2.777
Archaea	F	3.44	1.822
Archaea	G	7.97	2.708
Archaea	H	1.82	1.337
Archaea	I	7.73	2.671
Archaea	K	6.83	2.522

Group	AminoAcid	Percentage	StdErr
Archaea	L	8.78	2.83
Archaea	M	2.53	1.57
Archaea	N	3.59	1.861
Archaea	P	4.48	2.07
Archaea	Q	2.17	1.458
Archaea	R	5.58	2.295
Archaea	S	5.13	2.206
Archaea	T	4.74	2.125
Archaea	V	8.18	2.741
Archaea	W	0.78	0.88
Archaea	Y	3.2	1.761
Eukaryota	A	6.62	2.487
Eukaryota	C	1.77	1.32
Eukaryota	D	5.38	2.256
Eukaryota	E	6.64	2.489
Eukaryota	F	4.2	2.007
Eukaryota	G	6.19	2.41
Eukaryota	H	2.44	1.542
Eukaryota	I	5.44	2.269
Eukaryota	K	5.8	2.338
Eukaryota	L	10.1	3.013
Eukaryota	M	2.3	1.498
Eukaryota	N	4.47	2.067
Eukaryota	P	4.55	2.083
Eukaryota	Q	4.45	2.063
Eukaryota	R	5.35	2.25
Eukaryota	S	8.3	2.758
Eukaryota	T	5.39	2.258
Eukaryota	V	6.11	2.395
Eukaryota	W	1.33	1.148
Eukaryota	Y	3.16	1.749

We can observe that length of proteins differ significantly between kingdoms and model organisms. We also can observe that There is significant difference between kinndoms and our model organism (couple percentages changes acidicity drastically).

> Remark: I use word significant because our probe is big enough to obtain high p-value through wilcoxons tests.

Comparison between data bases

Length Distribution Comparison

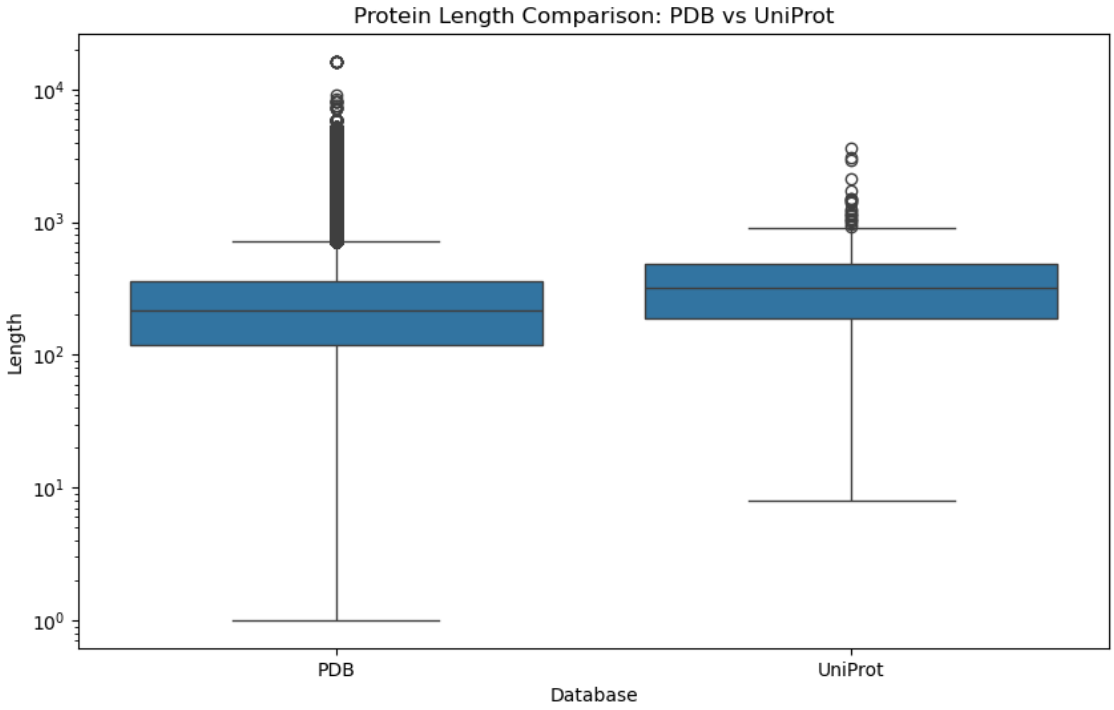


Figure 6: Length distribution comparison between PDB structures and UniProt sequences (log scale).

Amino Acid Composition Comparison

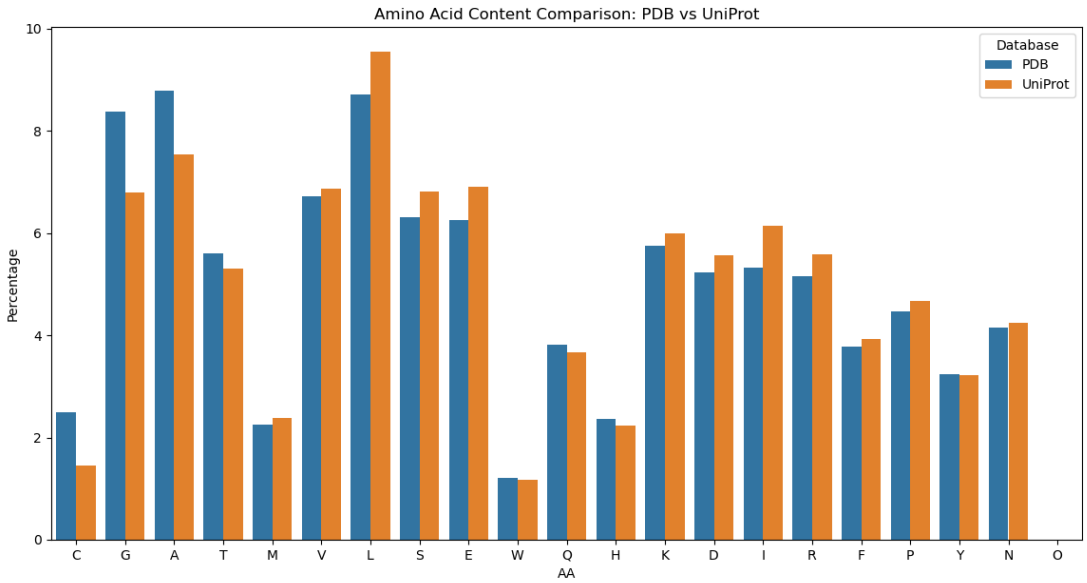


Figure 7: Average amino acid content comparison between PDB and UniProt.

Detailed Comparison Table

Group	Mean_Length	SE	Median_Length	Sample_Size
PDB	287.26	4.51	214	1104352
UniProt	391.35	18.63	319	400

AA Composition Stats (PDB vs UniProt)

Group	AminoAcid	Percentage	StdErr
PDB	A	8.78	0.027
PDB	C	2.5	0.015
PDB	D	5.23	0.021
PDB	E	6.25	0.023
PDB	F	3.78	0.018
PDB	G	8.37	0.026
PDB	H	2.36	0.014
PDB	I	5.33	0.021
PDB	K	5.76	0.022
PDB	L	8.72	0.027
PDB	M	2.25	0.014
PDB	N	4.16	0.019
PDB	P	4.47	0.02
PDB	Q	3.82	0.018
PDB	R	5.15	0.021
PDB	S	6.32	0.023
PDB	T	5.6	0.022
PDB	V	6.71	0.024
PDB	W	1.22	0.01
PDB	Y	3.23	0.017
UniProt	A	7.53	1.319
UniProt	C	1.44	0.596
UniProt	D	5.56	1.146
UniProt	E	6.91	1.268
UniProt	F	3.93	0.971
UniProt	G	6.8	1.259
UniProt	H	2.23	0.738
UniProt	I	6.14	1.2
UniProt	K	6	1.187
UniProt	L	9.55	1.47
UniProt	M	2.39	0.763
UniProt	N	4.25	1.008
UniProt	P	4.68	1.056
UniProt	Q	3.66	0.939
UniProt	R	5.59	1.148
UniProt	S	6.81	1.26
UniProt	T	5.3	1.12
UniProt	V	6.87	1.265
UniProt	W	1.17	0.538
UniProt	Y	3.22	0.882

We can observe huge difference between PDB (287 AA) and UniProt (391 AA) databases. That is because PDB, contains only experimentally determinted structures which overrepresented shorter structures. For this process proteins needed to be crystalized for X-ray difffraction, which is problematic with longer proteins. UniProt contains also predicted proteins which are longer. ## Bibliography

1. Kühlbrandt, W. (2014). The Resolution Revolution. *Science*, 343(6178), 1443-1444.
2. Varshavsky, A. (2011). The N-end rule pathway and regulation by proteolysis. *Protein Science*, 20(8), 1298-1345.